# Empirical Evaluation of Vision Transformers and ResNets in Computer Vision

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## 1. Abstract & Research Motivation

Convolutional Neural Networks (CNNs) have historically dominated visual recognition tasks due to their translational invariance and local receptive fields. Recently, Vision Transformers (ViT) have challenged this dominance by treating image patches as sequence tokens.

In this paper, we conduct an extensive empirical study comparing ResNet-50 and ViT-B/16 across ImageNet-1K classification, robustness to out-of-distribution noise, and inference throughput on edge devices.

## 2. Experimental Setup and Hyperparameters

Both models were trained using RandAugment data augmentation, CutMix, and Mixup over 300 epochs. Optimization used SGD with momentum 0.9 for ResNet, and AdamW with weight decay 0.05 for ViT.

Images were resized to 224x224 pixels with 16x16 patch projection for ViT-B/16. Total parameter count is 25.6M parameters for ResNet-50 compared to 86.4M parameters for ViT-B/16.

## 3. Comparative Results & Accuracy

On ImageNet-1K top-1 accuracy, ViT-B/16 achieved 81.2% while ResNet-50 achieved 79.1%. However, on corrupted ImageNet-C benchmarks, ViT demonstrated higher resilience with 14% lower error rate under severe adversarial Gaussian noise.

Conversely, inference speed on edge platforms (NVIDIA Jetson AGX Orin) favored ResNet-50, which executed at 142 frames per second (FPS), compared to 54 FPS for ViT-B/16 due to attention matrix latency.

## 4. Conclusions and Open Challenges

Our findings indicate that while Vision Transformers scale superiorly with massive pre-training data, CNNs remain optimal for constrained hardware deployments.

Future research should explore hybrid CNN-Transformer architectures that balance inductive bias with self-attention flexibility.